

PROGRESS

HelixCode CLI-Agent Fusion — Live Progress Tracker

STOP/RESUME PROTOCOL: read this file first. The “current focus” pointer below identifies the active task. The “evidence trail” links every claim of “done” to its commit + Challenge output.

Spec: docs/superpowers/specs/2026-05-04-cli-agent-fusion-synthesis-design.md
Plan: docs/superpowers/plans/2026-05-04-phase-0-foundation-cleanup.md

Current focus

- **Active phase:** Post-design — CONST-046 i18n campaign (Phase 4 active)
- **Active feature:** Systematic challenges/pkg/userflow/ migration (rounds 100+)
- **Active task:** Rounds 99a (panoptic) + 100 (evaluators.go) + 101 (ai_testgen.go) + 102 (challenge_desktop.go) running concurrently
- **Last completed:** Round 99b — CONST-046 audit-gate tightening with baseline (commit 3f4f110)
- **Owner:** agent (Claude Opus 4.7 1M-context)
- **Started:** 2026-05-04 (P0)
- **Last touched:** 2026-05-19 (round 99b close-out¹²⁶)
- **Blocked-on:** —

Phase status

Phase	State	Started	Completed	Evidence
P0 — Foundation	done	2026-05-04	2026-05-05	docs/improvements/05_phase_0_evidence.md
P1 — claude-code	done	2026-05-05	2026-05-06	docs/improvements/06_phase_1_evidence.md
P1.5 — Foundation Cleanup	done	2026-05-06	2026-05-06	docs/improvements/06_phase_1_evidence.md §P1.5
P2 — Other CLI agents	done	2026-05-06	2026-05-07	docs/improvements/07_phase_2_evidence.md
P3 — Test infra	done	2026-05-07	2026-05-08	docs/improvements/PROGRESS.md §Phase 3
P4 — Anti-bluff audit	done	2026-05-08	2026-05-08	docs/improvements/PROGRESS.md §Phase 4
P5 — End-user materials	done	2026-05-08	2026-05-08	this section

Phase	State	Started	Completed	Evidence
Round 37-89 — Governance + stabilization + LLM wiring	done	2026-05-08	2026-05-19	docs/ CONTINUATION.md close-outs ⁹⁴⁻¹¹⁶
CONST-046 Phase 1 (rounds 91-93) — pkg/ i18n core + audit + injection	done	2026-05-19	2026-05-19	docs/ CONTINUATION.md close-outs ¹¹⁸⁻¹²⁰ ; commits e29b075/57de105/03e131f
CONST-046 Phase 2 (rounds 94-96) — SelfImprove×8 + HelixLLM×2 + harmony_os×5	done	2026-05-19	2026-05-19	docs/ CONTINUATION.md close-outs ¹²¹⁻¹²³ ; commits a39d855/ abe0319/1eb1851
CONST-046 Phase 3 (rounds 97-99) — DocProcessor + Planning + VisionEngine + audit-gate	99b done; 99a/97/98 closed; formal reports pending	2026-05-19	(in progress)	docs/ CONTINUATION.md close-outs ¹²⁴⁻¹²⁶ ; commits e584e4b/6abed9b/ 2d0c35b/3f4f110
CONST-046 Phase 4 (rounds 100+) — challenges/pkg/ userflow/ systematic	active	2026-05-19	(in progress)	docs/ CONTINUATION.md (pending); rounds 100/101/102 concurrent

Authoritative round-by-round narrative: docs/CONTINUATION.md (CONST-044). This file complements with phase-level rollup.

Governance trackers (per Constitution §11.4.15-19 + CONST-057): - docs/Issues.md + docs/Issues_Summary.md — open items (8 open as of 2026-05-19; 3 BLOCKED on operator) - docs/Fixed.md + docs/Fixed_Summary.md — closure ledger (17 items closed in round-system) - All four kept in sync with this file via CONST-049 §11.4.17 commit discipline.

P1.5 Work-package list (12 WPs) — ALL CLOSED



P1.5-WP1 — Inventory + foundation safety (5 tasks) ← commit 421495a



P1.5-WP2 — Submodule restructuring (~67 mechanical moves) ← commit 90dec95



P1.5-WP3 — Submodule deduplication (5 sets) ← commit 154c06c



- ☒ P1.5-WP4 — API key loader (bash + Go) ← commit e57894e
- ☒ P1.5-WP5 — .env API key dedup (USER GATE) ← commit 92d5463
- ☒ P1.5-WP6 — Docs consolidation (3 dirs) ← commit f09f57d
- ☒ P1.5-WP7 — Snake_case directory normalization ← commit 3c3cd8d
- ☒ P1.5-WP8 — Anti-bluff Constitution propagation ← commit 0eead08
- ☒ P1.5-WP9 — Reference updates (comprehensive grep sweep) ← commit 42166fd
- ☒ P1.5-WP10 — Rebuild + validation + fix internal/tools/git pre-existing failure ← commit 0a77c93 + fix 45be827
- ☒ P1.5-WP11 — Phase 1.5 Challenge harness (5 phases) ← meta-repo 306d3d9 + Challenges submodule 7e94f28
- ☒ P1.5-WP12 — Commit + push everything (deepest-first) ← (this commit)

P1.5 task list (in flight)

- ☒ P1.5-pre — HelixAgent gitlink commit ← aad6a67d
- ☒ P1.5-pre — Challenges gitlink commit ← 47dc905a
- ☒ P1.5-pre — root tracked + gitlink commit ← d0ad6fd3
- ☒ P1.5-pre — remove Example_Projects/ entirely (67 submodules) ← ad5e108c
- ☒ P1.5-pre — gitignore phase-1 development artefacts ← cff2d90f
- [~] P1.5-WP1-T01.01 — recursive fetch + pull every submodule **HALTED** (init aborted at helix_agent/DebateOrchestrator unreachable; full log in docs/improvements/p1-5-fetch.log)
- [~] P1.5-WP1-T01.02 — pre-state snapshot **PARTIAL** — docs/improvements/p1-5-snapshot-pre.md (commit pending bootstrap)
- [~] P1.5-WP1-T01.03 — remote reachability sweep **IN PROGRESS** (181 unique URLs, see docs/improvements/p1-5-remote-reachability.md)
- ☒ P1.5-WP1-T01.04 — dedup canonical list ← docs/improvements/p1-5-dedup-canonical.md
- ☐ P1.5-WP1-T01.05 — bootstrap evidence + advance PROGRESS (commit pending — this commit)

Active phase task list (Phase 0)

- ☒ P0-01 — bootstrap PROGRESS.md ← commit 2c07f57

- [~] P0-02 — Agent-Deck nested-worktree recursion error: **DEFERRED** (cosmetic; safe-fix requires modifying third-party submodules which is out of scope per spec §2.1; original `.git/info/exclude` approach was based on incorrect understanding of git submodule recursion semantics; see parking lot). Reverts: commits 904c925 + a82f1a9.
- ☐ P0-03 — add HelixAgent submodule ← (this commit) — 47/60 cli_agents populated; 13 deferred to Phase 2 sub-specs (see parking lot)
- ☐ P0-04 — verify-llmsverifier-pin-parity.sh ← (this commit)
- ☐ P0-05 — migrate API keys from `../helix_agent/.env` ← (this commit)
- ☐ P0-06 — update `.gitignore` (root + inner) ← (this commit)
- ☐ P0-07 — refresh `helix_code/helix_code/.env.example` ← (this commit)
- ☐ P0-08 — scan-secrets.sh + planted-secret test ← (this commit)
- ☐ P0-08.5 — remediate 3 pre-existing tracked credentials ← commits 8d30add + 15cca9b + 45bd4d4
- ☐ P0-08.7 — Port SonarQube + Snyk security scan integration through containers ← commits 1d728de + 2494bc8 + e29e2f6 + 16a4490 + sub5; Challenges: 33/33 + 26/26 PASS; Go BootManager wiring landed (go build exits 0); live scans deferred pending credential rotation (see evidence §P0-08.7)
- ☐ P0-08.7-fix — T08.7 code-quality review findings (Critical 1+2, Important 3-7) ← commits b21b051; evidence: §P0-T08.7 (fix-it) in 05_phase_0_evidence.md
- ☐ P0-09 — pre-push hook + installer + setup.sh wiring ← (this commit)
- ☐ P0-10 — create `helix_code/{CLAUDE,AGENTS,CONSTITUTION}.md` (inner Go-app governance triplet) ← (this commit)
- ☐ P0-11 — add Article XII (CONST-042, CONST-043) to root `CONSTITUTION.md` ← (this commit)
- ☐ P0-12 — cascade CONST-042/043 + anti-bluff anchor to root sister files (CLAUDE, AGENTS, CRUSH, QWEN) ← (this commit)
- ☐ P0-13 — fix root `CLAUDE.md` §3.2 bluff (HelixCode tracked-dir vs. submodule) ← (this commit)
- ☐ P0-14 — extend `verify-governance-cascade.sh` + run `propagate-governance.sh` + cascade CONST-042/043 across owned-by-us submodules ← (this commit)
- ☐ P0-15 — Makefile `verify-foundation` target + extend `ci-validate-all` ← (this commit)
- ☐ P0-16 — regenerate diagrams + DEPRECATED.md pointers + Phase 0 evidence + push close-out ← (this commit)

Active feature task list (P1-F01: Auto-Compaction)



- ☐ P1-F01-T01 — bootstrap Phase 1 evidence + advance PROGRESS ← commit f0b9b15
- ☐ P1-F01-T02 — add GetContextWindow + CountTokens to Provider interface ← commit 5b153e6
- ☐ P1-F01-T03 — implement Provider methods across all *_provider.go ← commit 827971f
- ☐ P1-F01-T04 — ThrashingGuard with TDD ← commit 59f7daa
- ☐ P1-F01-T05 — CompactionMetadata with TDD ← commit b9eae7f
- ☐ P1-F01-T06 — AutoCompactor with TDD ← commit 4330341
- ☐ P1-F01-T07 — wire AutoCompactor into internal/agent ← commit cace643
- ☐ P1-F01-T08 — wire ThrashingGuard reset into internal/session/manager.go ← commit b913ce2
- ☐ P1-F01-T09 — integration test against real Anthropic provider ← commit 4734f35
- ☐ P1-F01-T10 — Challenge with expected.json + runtime evidence ← commit 9284392
- ☐ P1-F01-T11 — Feature 1 close-out + push ← (this commit)

Active feature task list (P1-F02: Permission Rule System)

- ☐ P1-F02-T01 — bootstrap evidence + advance PROGRESS ← commit d56905d
- ☐ P1-F02-T02 — add Wildcard field to confirmation.Condition (TDD) ← commit 5ffc46d
- ☐ P1-F02-T03 — internal/tools/permissions package skeleton ← commit 26de1b4
- ☐ P1-F02-T04 — shell_splitter.go + mvdan.cc/sh/v3 dep (TDD) ← commits 28a4fa8 + c2b5dd8
- ☐ P1-F02-T05 — rule_engine.go pattern parse + match + priority (TDD) ← commit eab41d3
- ☐ P1-F02-T06 — mode_presets.go five presets + command lists (TDD) ← commit 75b284f
- ☐ P1-F02-T07 — rule_loader.go YAML + file precedence (TDD) ← commit 31c4366
- ☐ P1-F02-T08 — permissions.go facade + PolicyEngine registration ← commit 41be967

- ☐ P1-F02-T09 — wire `-permission-mode` flag + integration test (no mocks) ← commit `c1d67ad`
- ☐ P1-F02-T10 — helixcode permissions `{list,add,remove,check}` subcommands ← commit `588f2cd`
- ☐ P1-F02-T11 — `/permissions` slash command via `internal/commands` ← commits `2fb11d4` + `244aff9`
- ☐ P1-F02-T12 — Challenge with three runtime-evidence scenarios ← commit `7252911`
- ☐ P1-F02-T13 — Feature 2 close-out + push ← (this commit)

Active feature task list (P1-F03: Tool Result Persistence)

- ☐ P1-F03-T01 — bootstrap evidence + advance PROGRESS — `ee35017`
- ☐ P1-F03-T02 — `internal/tools/persistence` package skeleton (types + doc) — `c806f72`
- ☐ P1-F03-T03 — `Manager.MaybePersist` with hash idempotence (TDD) — `38a17d4`
- ☐ P1-F03-T04 — `LoadPersisted` with path-traversal guard (TDD) — `a9a41f2`
- ☐ P1-F03-T05 — `CleanupOld` with filename-pattern matching (TDD) — `7afe24f`
- ☐ P1-F03-T06 — wire into `internal/llm/tool_provider.go` orchestration loop — `6199e96`
- ☐ P1-F03-T07 — audit + wire individual LLM providers — `88856c4`
- ☐ P1-F03-T08 — system prompt note about `persistedOutputPath` — `c80b438`
- ☐ P1-F03-T09 — `cmd/cli/main.go` startup + integration test (no mocks) — `9141297`
- ☐ P1-F03-T10 — Challenge with three runtime-evidence scenarios — `84874be`
- ☐ P1-F03-T11 — Feature 3 close-out + push — `8b13e93`

Active feature task list (P1-F04: Git Worktree Agent Isolation)

- ☐ P1-F04-T01 — bootstrap evidence + advance PROGRESS + `.gitignore` ← commit `d5ae14a`
- ☐ P1-F04-T02 — `internal/tools/worktree` package skeleton (types + doc) ← commits `97075a2` + `ccaaf33`



P1-F04-T03 — git.go thin git-binary wrappers (TDD against ephemeral repo) ← commit 3e8b942



P1-F04-T04 — Manager + ValidateName + GetCurrentDirectory + IsIsolated (TDD) ← commit 94dec d8



P1-F04-T05 — Manager.EnterWorktree (TDD; existing/new branch + dirty rejection) ← commit bdd f79d



P1-F04-T06 — Manager.ExitWorktree + ListWorktrees + RemoveWorktree (TDD) ← commit 1 f a0617



P1-F04-T07 — 4 tools.Tool interface implementations (TDD) ← commit f522805



P1-F04-T08 — session.Manager currentWorktree field + getter/setter (TDD) ← commit 73b040a



P1-F04-T09 — helixcode worktree {list,enter,exit,remove} Cobra subcommands ← commit 0a1fb53



P1-F04-T10 — /worktree slash command + register in builtin/register.go ← commit 64e8a45



P1-F04-T11 — cmd/cli/main.go startup wiring + integration test (no mocks) ← commit 4325459



P1-F04-T12 — Challenge with three runtime-evidence scenarios ← commit 9a23db1



P1-F04-T13 — Feature 4 close-out + push ← (this commit)

Active feature task list (P1-F05: Hook-Based Extensibility)



P1-F05-T01 — bootstrap evidence + advance PROGRESS ← commit b7e7185



P1-F05-T02 — add 6 new HookType constants (TDD) ← commit 857ef64



P1-F05-T03 — yaml_loader.go: FileLoader + apiVersion validation (TDD) ← commits bf50e8d + df72487



P1-F05-T04 — shell_runner.go: NewShellRunner HookFunc (TDD) ← commits af5641f + b304c3e



P1-F05-T05 — blockers.go: Blockers helper (TDD) ← commit b820bee



P1-F05-T06 — wire registry.Execute with 6 events (TDD) ← commit 61ce79e



- ☐ P1-F05-T07 — wire OnCompaction in AutoCompactor (TDD) ← commit 302aabd
- ☐ P1-F05-T08 — wire OnError + RequestPlanApproval stub in agent.go (TDD) ← commit 76a0823
- ☐ P1-F05-T09 — helixcode hooks {list,test,enable,disable,validate} subcommands ← commit d0f85d9
- ☐ P1-F05-T10 — /hooks slash command + builtin registration ← commit 910488b
- ☐ P1-F05-T11 — cmd/cli/main.go startup wiring + integration tests (no mocks) ← commit 6925038
- ☐ P1-F05-T12 — Challenge with three runtime-evidence scenarios ← commit d5da040
- ☐ P1-F05-T13 — Feature 5 close-out + push ← (this commit)

Active feature task list (P1-F06: MCP Full Lifecycle)

- ☐ P1-F06-T01 — bootstrap evidence + advance PROGRESS
- ☐ P1-F06-T02 — types.go + transport.go interface + BackoffSchedule (TDD)
- ☐ P1-F06-T03 — transport_stdio.go + cross-platform unix/windows files (TDD)
- ☐ P1-F06-T04 — transport_http.go with OAuth bearer header (TDD)
- ☐ P1-F06-T05 — transport_sse.go with reconnect loop (TDD)
- ☐ P1-F06-T06 — transport_ws.go via gorilla/websocket (TDD)
- ☐ P1-F06-T07 — oauth.go: RFC 8414 discovery + PKCE + token cache (TDD)
- ☐ P1-F06-T08 — lifecycle.go: Client state machine + handshake (TDD)
- ☐ P1-F06-T09 — registry.go: Manager + tool merging (TDD)
- ☐ P1-F06-T10 — config.go: YAML loader/saver, project + user precedence (TDD)
- ☐ P1-F06-T11 — cmd/cli/mcp_cmd.go + /mcp slash command (TDD)
- ☐ P1-F06-T12 — cmd/cli/main.go startup wiring + tools/registry.go integration + integration tests
- ☐ P1-F06-T13 — Challenge with runtime evidence + cross-compile check
- ☐ P1-F06-T14 — Feature 6 close-out + push

Active feature task list (P1-F07: Background Task System)



P1-F07-T01 — bootstrap evidence + advance PROGRESS



P1-F07-T02 — workflow/background.go: BackgroundTask + TaskState (TDD)



P1-F07-T03 — workflow/background.go: BackgroundManager + sweeper (TDD)



P1-F07-T04 — tools/types_background.go: BackgroundAware interface + LineSink + error sentinel



P1-F07-T05 — tools/shell/background.go: shell BackgroundAware adapter (TDD)



P1-F07-T06 — tools/registry.go: SetBackgroundManager + Execute dispatch + adaptToolForBackground (TDD)



P1-F07-T07 — tools/task_tools.go: TaskOutputTool + TaskStopTool + RegisterTaskTools (TDD)



P1-F07-T08 — commands/tasks_command.go: /tasks slash command + builtin registration (TDD)



P1-F07-T09 — cmd/cli/main.go startup wiring + integration test (real subprocess)



P1-F07-T10 — Challenge with runtime evidence + cross-compile check



P1-F07-T11 — Feature 7 close-out + push

Active feature task list (P1-F08: Plan Mode)



P1-F08-T01 — bootstrap evidence + advance PROGRESS



P1-F08-T02 — Planner extensions + hook types (TDD)



P1-F08-T03 — planmode/gating.go: ToolGate (TDD)



P1-F08-T04 — tools/types_planmode.go + plan_tools.go (TDD)



P1-F08-T05 — tools/registry.go: SetPlanModeGate + Execute gating (TDD)



P1-F08-T06 — commands/plan_command.go: /plan slash + builtin reg (TDD)



P1-F08-T07 — cmd/cli/main.go startup wiring + integration test



P1-F08-T08 — Challenge with runtime evidence + cross-compile check



P1-F08-T09 — Feature 8 close-out + push

Active feature task list (P1-F09: Slash Command System)



P1-F09-T01 — bootstrap evidence + advance PROGRESS



P1-F09-T02 — markdown_commands.go: MarkdownCommand + parser + substitution (TDD)



P1-F09-T03 — MarkdownLoader: scan dirs + register/unregister (TDD)



P1-F09-T04 — markdown_watcher.go: fsnotify + debounce (TDD)



P1-F09-T05 — /commands slash + helixcode commands cobra (TDD)



P1-F09-T06 — main.go wiring + integration test



P1-F09-T07 — Challenge with runtime evidence + cross-compile check



P1-F09-T08 — Feature 9 close-out + push

Active feature task list (P1-F10: Skill System)



P1-F10-T01 — bootstrap evidence + advance PROGRESS



P1-F10-T02 — markdown_skills.go: Skill + SkillRegistry + parser + Render (TDD)



P1-F10-T03 — SkillLoader: scan dirs + register/unregister (TDD)



P1-F10-T04 — skills_watcher.go: fsnotify + debounce (TDD)



P1-F10-T05 — agent/skill_dispatcher.go: Match + capture extraction (TDD)



P1-F10-T06 — /skills slash + helixcode skills cobra (TDD)



P1-F10-T07 — main.go wiring + integration test



P1-F10-T08 — Challenge with runtime evidence + cross-compile check



P1-F10-T09 — Feature 10 close-out + push

Active feature task list (P1-F11: Session Transcript Resume)



- ☒ P1-F11-T01 — bootstrap evidence + advance PROGRESS ← commit ddb45dc
- ☒ P1-F11-T02 — identity.go: ComputeProjectIdentity (TDD) ← commit fa6bc5f
- ☒ P1-F11-T03 — transcript_store.go: JSONL append/read + metadata I/O (TDD) ← commit 466ab97
- ☒ P1-F11-T04 — resume.go: ResumeFinder + ResumeMode + FindResumeTarget (TDD) ← commit d72e401
- ☒ P1-F11-T05 — SessionManager extensions: Append/Resume/CurrentID (TDD) ← commit 08fa5c0
- ☒ P1-F11-T06 — /sessions slash + helixcode sessions cobra (TDD) ← commit 607206a
- ☒ P1-F11-T07 — main.go -resume/-continue flag parsing + integration test ← commit 0fb036c
- ☒ P1-F11-T08 — Challenge with runtime evidence + cross-compile check ← submodule 1e79453 + meta-repo f4d0ff2
- ☒ P1-F11-T09 — Feature 11 close-out + push ← (this commit)
- ☒ P1-F11-fix — preserve ProjectPath/Name across SessionManager.Append ← commit f258cf7

Active feature task list (P1-F12: Multi-Provider Backend)

- ☒ P1-F12-T01 — bootstrap evidence + advance PROGRESS ← commit bd5dc69
- ☒ P1-F12-T02 — provider.go: unified interface + LLMsVerifier audit (TDD) ← commit 06c9c34
- ☒ P1-F12-T03 — anthropic_provider.go conformance + base URL precedence (TDD) ← commit dde10dd
- ☒ P1-F12-T04 — bedrock_provider.go conformance + verifier GetModels (TDD) ← commit d01026d
- ☒ P1-F12-T05 — vertexai_provider.go conformance + verifier GetModels (TDD) ← commit 67417ed
- ☒ P1-F12-T06 — azure_provider.go conformance + verifier GetModels (TDD) ← commit 880b4dc
- ☒ P1-F12-T07 — provider_factory.go: NewCloudProvider + Selector (TDD) ← commit 28b6fa1
- ☒

P1-F12-T08 — wizard.go: tview TUI + mode 0600 + O_EXCL (TDD) ← commit 778040e



P1-F12-T09 — main.go wiring + helixcode wizard cobra + integration test ← commit ac55fca



P1-F12-T10 — Challenge with runtime evidence (local + cloud-gated) ← submodule 4e42fbc + meta-repo b937e17 + SHA backfill 1586624



P1-F12-T11 — Feature 12 close-out + push 4 remotes ← (this commit)

Active feature task list (P1-F13: LSP Integration)



P1-F13-T01 — bootstrap evidence + advance PROGRESS to F13 ← commit df98b6d



P1-F13-T02 — go.mod: add go.lsp.dev/jsonrpc2 v0.10.0 + protocol v0.12.0 (TDD) ← commit b9a30e4



P1-F13-T03 — internal/tools/lsp_types.go: Diagnostic + DiagnosticSummary + LSPServerSpec (TDD) ← commit 3c5d894



P1-F13-T04 — internal/tools/lsp_client.go: jsonrpc2 wrapper + handshake (TDD with paired pipes) ← commit 2fdb648



P1-F13-T05 — internal/tools/lsp_manager.go: lazy-spawn + idle-timeout + ext-router + fake LSP (TDD) ← commit beef346



P1-F13-T06 — internal/tools/lsp_servers.go: curated 5-server allowlist + Detect (TDD) ← commit 33387a3



P1-F13-T07 — internal/tools/lsp.go: LSPGetDiagnostics + LSPAnalyzeDiagnostic tools (TDD) ← commit 9bb3118



P1-F13-T08 — registry.SetLSPManager + post-Execute auto-trigger for fs_edit/fs_write/multi_edit_commit (TDD) ← commit a1aa7e6



P1-F13-T09 — /lsp slash command (status/restart/list-servers/stop) (TDD) ← commit 1b7812f



P1-F13-T10 — helixcode lsp cobra + main.go wiring + gated integration test ← commit 080b79b



P1-F13-T11 — Challenge: in-tree fake LSP pipeline + gated real-server phase ← submodule f00bf19 + meta-repo 9ea2cdf



P1-F13-T12 — Feature 13 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F14: Sandboxed Shell Execution)



P1-F14-T01 — bootstrap evidence + advance PROGRESS to F14 ← commit 0ef5811



P1-F14-T02 — sandbox/types.go: SandboxConfig + Policy + Capabilities + Result + Backend interface + ConstitutionalDenylist (TDD) ← commit abdbdab



P1-F14-T03 — sandbox/detector.go: capability probes + SelectBackend with fail-closed (TDD) ← commit 4f7141f



P1-F14-T04 — sandbox/bubblewrap_backend.go: deterministic argv builder + Run (TDD) ← commit ec4cb9b



P1-F14-T05 — sandbox/native_backend.go: SysProcAttr.Cloneflags userns + native_helper re-exec (TDD) ← commit 5d05b3d



P1-F14-T06 — sandbox/manager.go: backend selection + CONST-033 deny + user deny + fail-closed (TDD) ← commit a642101



P1-F14-T07 — sandbox/sandboxed_shell_tool.go: Tool interface impl as shell_sandboxed (TDD) ← commit ba54c0c



P1-F14-T08 — sandbox/config_loader.go: YAML loader + secret-safe writer (mode 0600) (TDD) ← commit 9aad02



P1-F14-T09 — /sandbox slash command (status/test/policy) (TDD) ← commit 93dc377



P1-F14-T10 — main.go wiring (Detector + Manager + tool + slash) + gated integration test ← commit fdb5ddc



P1-F14-T11 — Challenge harness: detector + fail-closed always-runs + bwrap/native gated phases ← submodule 7d336ad + meta-repo 998896c



P1-F14-T12 — Feature 14 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F15: Subagent Team)



P1-F15-T01 — bootstrap evidence + advance PROGRESS to F15 ← commit b970aa5



P1-F15-T02 — subagent types + Isolation/State enums + FakeLLMProvider TEST PROVIDER (TDD) ← commit adc273d



P1-F15-T03 — InProcessSpawner with real llm.Provider invocation + ctx cancel (TDD) ← commit ceeb670



- ☐ P1-F15-T04 — SubprocessSpawner with sentinel env var + JSON stdout decode (TDD) ← commit e c 2 1 b 1 7
- ☐ P1-F15-T05 — SubagentManager with streaming dispatch + max-concurrency + kill-by-id (TDD) ← commit 8 e 2 f 9 e 8
- ☐ P1-F15-T06 — F04 worktree integration for isolation=worktree (real git tempdir test) ← commit 9 3 1 1 6 9 2
- ☐ P1-F15-T07 — TaskTool implementing tools.Tool as t a s k (claude-code-compatible name) ← commit 1 f 9 d 0 f 3
- ☐ P1-F15-T08 — IsSubagentInvocation + RunAsSubagent helper-mode + main.go early-dispatch ← commit 0 7 8 6 3 d 2
- ☐ P1-F15-T09 — /subagents slash command (list/status/kill) + CONST-042 anti-leak (TDD) ← commit 8 7 b 6 e a c
- ☐ P1-F15-T10 — wire SubagentManager into main.go + /subagents + gated integration tests ← commit a f 0 a a 2 9
- ☐ P1-F15-T11 — Challenge with runtime evidence (in-process + subprocess always; worktree + real-LLM gated) ← submodule 1 6 3 9 6 5 e + meta-repo 1 6 7 0 8 a 7
- ☐ P1-F15-T12 — Feature 15 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F16: OpenTelemetry Integration)

- ☐ P1-F16-T01 — bootstrap evidence + advance PROGRESS to F16 ← commit 5 f c 7 d c 1
- ☐ P1-F16-T02 — go.mod: add OTel v1.30.0 dep set (TDD failing import test) ← commit f 2 e 7 2 6 0
- ☐ P1-F16-T03 — types.go: TelemetryConfig + ExporterKind + DefaultBlockedAttributeKeys + sentinels (TDD) ← commit d e 9 4 1 b 4
- ☐ P1-F16-T04 — config.go + attribute_filter.go: env-var parsing + exporter selection + secret filter (TDD) ← commit 3 c 8 5 9 3 c
- ☐ P1-F16-T05 — provider.go: TelemetryProvider construction (TracerProvider + MeterProvider) (TDD) ← commit a 8 e 1 3 e 3
- ☐ P1-F16-T06 — llm_instrumentation.go: TracedLLMProvider decorator with token counter + latency histogram (TDD) ← commit 6 f c b f f 6
- ☐ P1-F16-T07 — tool_instrumentation.go + ToolRegistry.Execute in-place wrap + SetTelemetryProvider (TDD) ← commit d 8 0 c 2 7 8
- ☐

P1-F16-T08 — agent_instrumentation.go + BaseAgent.executeTaskWithLLM in-place wrap (TDD) ← commit 7c06806



P1-F16-T09 — /telemetry slash command (status/show/flush) (TDD) ← commit 7701c33



P1-F16-T10 — main.go wiring + gated integration tests (stdout always; OTLP gRPC+HTTP gated) ← commit a5eb1c9



P1-F16-T11 — Challenge harness: in-tree fake OTLP/HTTP receiver + 5 phases (STDOUT/FAKE-OTLP/FILTER/NOOP/REAL) ← submodule af34a2c + meta-repo c4972dc



P1-F16-T12 — Feature 16 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F17: Smart File Editing)



P1-F17-T01 — bootstrap evidence + advance PROGRESS to F17 ← commit 37b1471



P1-F17-T02 — smartedit/types.go: EditBlock + markers + sentinels + size limits (TDD) ← commit adcd9f0



P1-F17-T03 — smartedit/parser.go: SEARCH/REPLACE block parser + path-stickiness + line tracking (TDD) ← commit 91d7550



P1-F17-T04 — smartedit/applier.go + binary_detect.go: lenient re-search + ambiguity + binary refusal (TDD) ← commit 37beb27



P1-F17-T05 — smartedit/diff.go: unified-diff wrapper re-using F08 multiedit DiffManager (TDD) ← commit 6c00471



P1-F17-T06 — smartedit/smart_edit_tool.go: Tool impl + multiedit transaction + post-write re-read + diff (TDD) ← commit 721fed9



P1-F17-T07 — /edit slash (status/diff/dry-run/commit) + SmartEditInspector (TDD) ← commit a2dd7eb



P1-F17-T08 — main.go wiring + registry registration + always-run integration tests ← commit 5bf4c92



P1-F17-T09 — Challenge harness: 7-phase (SINGLE/NOT-FOUND/MULTI/ROLLBACK/DIFF/AMBIG/BINARY) with sha-256 positive evidence ← submodule e2e9e94 + meta-repo daa1279



P1-F17-T10 — Feature 17 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F18: No-Flicker Rendering)



- P1-F18-T01 — bootstrap evidence + advance PROGRESS to F18 ← commit a8aa8f3
- P1-F18-T02 — render/types.go: Renderer interface + RenderMode + Frame + sentinels + env-var consts (TDD) ← commit 8d6ec3b
- P1-F18-T03 — render/ansi_renderer.go: in-place line update + multi-line frame + dirty-region diff (TDD) ← commit a3cd0e1
- P1-F18-T04 — render/plain_renderer.go: line-by-line fallback with zero-ANSI/zero-CR invariant (TDD) ← commit 487d72e
- P1-F18-T05 — render/viewport.go: Frame buffer + dirty-line tracking + pure-Go Diff (TDD) ← commit 8c90e7c
- P1-F18-T06 — render/factory.go: HELIXCODE_RENDER env var + TTY detection via x/term (TDD) ← commit 288e6cd
- P1-F18-T07 — Wire LLM streaming hook in cmd/cli/main.go::handleGenerate (TDD) ← commit 4ece7e8
- P1-F18-T08 — render/tool_helpers.go + wire tool-result frame rendering (TDD) ← commit 05434c4
- P1-F18-T09 — Challenge harness: 5 phases (STREAMING-FANCY + STREAMING-PLAIN + DIRTY-REGION-DIFF + TTY-FALLBACK + REAL-TTY gated) ← submodule c409ed3 + meta-repo c44b049
- P1-F18-T10 — Feature 18 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F19: AskUserQuestion with Previews)

- P1-F19-T01 — bootstrap evidence + advance PROGRESS to F19 ← commit b7fc6bb
- P1-F19-T02 — askuser/types.go: Choice + Question + Result + Prompter interface + sentinels (TDD) ← commit 3be8ca7
- P1-F19-T03 — askuser/stdin_prompter.go: non-TTY short-circuit + retry loop + timeout + F18 menu render (TDD) ← commit 2ded9d2
- P1-F19-T04 — askuser/ask_user_tool.go: AskUserTool wrapping Prompter + CategoryAskUser (TDD) ← commit bc84789
- P1-F19-T05 — wire ask_user into registry + integration test (always-runs both branches) ← commit 98bd424
- P1-F19-T06 — Challenge harness: 5 always-run phases + reader-position + byte-offset positive evidence ← submodule e3454a4 + meta-repo ec b3e26



P1-F19-T07 — Feature 19 close-out + push 4 remotes non-force ← (this commit)

Active feature task list (P1-F20: Theme System) — FINAL Phase 1 feature



P1-F20-T01 — bootstrap evidence + advance PROGRESS to F20 ← commit 60777fd



P1-F20-T02 — theme/types.go: Role + Color + ColorDepth + Theme + sentinels + Reset const (TDD) ← commit 4697565



P1-F20-T03 — theme/builtin.go: dark/light/none themes with pinned byte tables (TDD) ← commit a66737d



P1-F20-T04 — theme/detect.go: ThemeName + ColorDepth detection from injected env (TDD) ← commit 7f97f57



P1-F20-T05 — theme/loader.go: ThemeRegistry + YAML merge into dark baseline + Styler (TDD) ← commit 1fd42d9



P1-F20-T06 — wire theme.Styler into handleGenerate via F18 RenderTextBlock (TDD) ← commit 1066798



P1-F20-T07 — /theme slash (status/list/show) + main.go wiring + integration test (TDD) ← commit 348630c



P1-F20-T08 — Challenge harness: 5 always-run phases (BUILT-IN-DARK + BUILT-IN-LIGHT + PLAIN-ZERO-COLOR + DEPTH-DETECT + YAML-MERGE) ← submodule 4bf04bb + meta-repo 300f973



P1-F20-T09 — Feature 20 close-out + push 4 remotes — PHASE 1 COMPLETE ← (this commit)

P2-F21 task list (Codex Approval Modes) — ALL CLOSED



P2-F21-T01 — bootstrap Phase 2 evidence + advance PROGRESS to F21 ← commit a7a349f



P2-F21-T02 — approval/types.go: ApprovalMode + ApprovalLevel + Decision + sentinels + ModeDescriptors (TDD) ← commit 9c2664d



P2-F21-T03 — approval/selector.go: flag > env > config > default precedence (TDD) ← commit 0d655d8



P2-F21-T04 — approval/manager.go: ApprovalManager with 4×4 matrix gate + F02/F14/F19 integration (TDD) ← commit 5ef13b8



P2-F21-T05 — Extend Tool interface with RequiresApproval() + DefaultLevelEdit + apply to ~30 existing tools (TDD) ← commit 19b f f c e + CONTINUATION update 1195e f 9



P2-F21-T06 — /approval slash command (status/set/show) (TDD) ← commit ad8843b + CONTINUATION update 9b72c26



P2-F21-T07 — main.go wiring + -approval pflag + registry hook + integration test (TDD) ← commit c022968 + CONTINUATION update bd67324



P2-F21-T08 — Challenge harness 5 phases (suggest-deny + auto-edit-prompt + full-auto-sandbox + runtime-change + F02-final-deny) ← Challenges submodule f2ea964 + meta-repo 2781c1a + CONTINUATION update ee413c3



P2-F21-T09 — Feature 21 close-out + push 4 remotes non-force ← (this commit)

P2-F22 task list (Aider Git Auto-Commit Per Change) — ALL CLOSED



P2-F22-T01 — bootstrap F22 evidence section + advance PROGRESS to F22 ← commit 550be34



P2-F22-T02 — autocommit/types.go: CommitContext + CommitResult + Options + sentinels + EnvVarName/CoAuthorTrailer/SkipParamKey constants (TDD) ← commit 0468beb



P2-F22-T03 — autocommit/git.go: thin git wrapper (real-git TDD) ← commit cb4fc30



P2-F22-T04 — autocommit/summariser.go + secret_filter.go (TDD) ← commit 4b2ab67



P2-F22-T05 — autocommit/committer.go: AutoCommitter.MaybeCommit pipeline + atomic.Bool enabled + co-author trailer (real-git TDD) ← commit 3a28ca6



P2-F22-T06 — registry.go: SetAutoCommitter + post-Execute fireAutoCommit hook + per-tool mutated-paths derivation (TDD) ← commit db55e72



P2-F22-T07 — /git_auto_commit slash command (status/on/off/show) (TDD) ← commit a999b3a



P2-F22-T08 — main.go wiring + integration test ← commit bab7ebc



P2-F22-T09 — Challenge harness 6+1 phases + close-out + push 4 remotes ← (this commit)

P2-F23 task list (Cline Browser Tool) — ALL CLOSED



P2-F23-T01 — bootstrap F23 evidence section + advance PROGRESS to F23 ← commit 64e499b



P2-F23-T02 — browser/types.go + options.go: Snapshot + ScreenshotResult + ManagerStatus + sentinels + EnvVarHeadedMode + MaxSnapshotBytes + MaxScreenshotBytes (TDD) ← commit cdb323e



P2-F23-T03 — browser/manager.go + session.go: BrowserManager atomic-pointer lifecycle + BrowserSession sync.Once close + sessionFactory seam (TDD) ← commit e0ff1bf



P2-F23-T04 — browser_navigate_v2.go: browser_navigate Tool impl with WaitReady + lazy session-create + 30s timeout (TDD); legacy tools renamed to browser_legacy_* ← commit 2bcb281



P2-F23-T05 — browser_snapshot_v2.go: browser_snapshot Tool impl with html/text mode + 64KB cap + Truncated flag (TDD) ← commit ec0b3cc



P2-F23-T06 — browser_click_type_v2.go: browser_click + browser_type Tool impls with NodeVisible + 5s + ClickWait settle (TDD) ← commit 659307d



P2-F23-T07 — browser_screenshot_v2.go: PNG-magic + DecodeConfig + size>1024 + tempdir 0600 (TDD) ← commit 3ca50a3



P2-F23-T08 — browser_close_v2.go: idempotent CloseSession + post-close RequireSession fails (TDD) ← commit ad0c0df



P2-F23-T09 — /browser slash + main.go wiring + RegisterBrowserToolsV2 + integration test (7/7 PASS against real chromium) ← commit f39f686



P2-F23-T10 — Challenge harness 7 phases A-G (chromium-gated) + close-out + push 4 remotes non-force ← (this commit)

P2-F24 task list (Codex Project Memory) — ALL CLOSED



P2-F24-T01 — bootstrap F24 evidence section + advance PROGRESS to F24



P2-F24-T02 — projectmemory/types.go: Memory + Render + sentinels + MaxMemoryBytes + DiscoveryFilenames (TDD)



P2-F24-T03 — projectmemory/loader.go: parent-walk + git-root-stop + user overlay + truncation (TDD with real tempdirs)



P2-F24-T04 — projectmemory/registry.go: atomic-pointer Snapshot/Set/Reload + MemorySnapshotter (TDD -race)



P2-F24-T05 — projectmemory/watcher.go: fsnotify + 200ms debounce + graceful degrade (TDD with real fsnotify)



P2-F24-T06 — /memory slash command (status/show/edit/reload) + editor seam (TDD)



P2-F24-T07 — BaseAgent SetMemoryRegistry + main.go wiring + integration test (TDD)



P2-F24-T08 — Challenge harness 5 phases A-E + close-out + push 4 remotes non-force

P2-F25 task list (Plandex Plan Trees) — ALL CLOSED



P2-F25-T01 — bootstrap F25 evidence section + advance PROGRESS to F25 — c744a27



P2-F25-T02 — plantree/types.go: PlanNode + PlanTree + PlanStatus + sentinels + constants + RenderTree (TDD) — 1edc117



P2-F25-T03 — plantree/store.go: FileStore Save/Load/List/Delete + atomic writes + Store interface (TDD real-tempdirs) — 889126d



P2-F25-T04 — plantree/operations.go: CreateTree + BranchNode + MergeNode pure-tree transforms (TDD) — 32a4139



P2-F25-T05 — plantree/verify.go: VerifyTree 6 checks + cycle-safe traversal (TDD) — 44010e5



P2-F25-T06 — plantree/compact.go: CompactTree + Summariser adapter + threshold guard (TDD) — 684e702



P2-F25-T07 — plantree/plan_tools.go: Six tools.Tool implementations (TDD) — ee054db



P2-F25-T08 — /plantree slash command (list/show/compact/verify) (TDD) — b7de19d



P2-F25-T09 — main.go wiring + plantree.RegisterPlanTools + /plantree registration + build verify — b716a07



P2-F25-T10 — Challenge harness 7 phases A-G + close-out + push 4 remotes non-force — ff9097d

P2-F26 task list (Openhands Workspace + Planner) — ALL CLOSED



P2-F26-T01 — bootstrap F26 evidence section + advance PROGRESS to F26 — 613b204



P2-F26-T02 — workspace/types.go + manager.go (TDD) — 975de95



- ☐ P2-F26-T03 — workspace/workspace_tools.go (TDD) — 4eb174d
- ☐ P2-F26-T04+T05 — planner types + executor + tools (TDD) — 9254a68
- ☐ P2-F26-T06+T07 — /openhands slash + main.go wiring + build verify — 5cdc6e7
- ☐ P2-F26-T08 — Challenge harness 5 phases A-E + close-out — b7572c0

P2-F27 task list (Aider Voice + Repo-Map) — ALL CLOSED

- ☐ P2-F27-T01 — bootstrap F27 evidence + advance PROGRESS — 0dc01b3
- ☐ P2-F27-T02-T07 — voice + repomap + /aider slash + main.go wiring — 29218cc
- ☐ P2-F27-T09 — Challenge harness 6 phases A-F + close-out — (this commit)

Decision log

- 2026-05-04 — Approach A (HelixAgent as integration substrate) — user-approved during brainstorming — see synthesis spec §2.1
- 2026-05-04 — Non-force pushes pre-authorised for the duration of this programme — user statement during brainstorming — see synthesis spec §7.3
- 2026-05-04 — claude-code-source is Phase 1 priority #1 — user statement — see synthesis spec §4.1
- 2026-05-05 — Phase 0 closed; 17 plan tasks done + 2 added during execution (T08.5, T08.7); foundation verified; carry-forward items documented in evidence file P0-16
- 2026-05-05 — Phase 1 entered; Feature 1 (Auto-Compaction) starts. Approach: extend existing internal/llm/compression infrastructure rather than build the parallel system the porting doc proposed (gap discovered during plan-writing).
- 2026-05-05 — Feature 1 (Auto-Compaction) closed. Eleven sub-commits; extended existing internal/llm/compression rather than building parallel infrastructure as the porting doc proposed. Per-provider native tokenizers deferred to Phase 3.
- 2026-05-05 — Feature 2 (Permission Rule System) closed. Thirteen+ sub-commits (T11 needed a registration follow-up 244aff9). Extended internal/tools/confirmation.PolicyEngine with a Wildcard Condition field; added internal/tools/permissions package that loads layered YAML rule files (~/.helixcode + project) and produces a Policy that delegates to a smuggle-resistant rule engine (mvdan.cc/sh/v3 walker handles \$(...), backticks, heredocs, quoted operators, pipelines). Five claude-code mode presets (default | auto | acceptEdits | dontAsk | bypassPermissions) compose with the existing AutonomyMode gradient. Full CLI surface: -permission-mode flag, helixcode permissions {list,add,remove,check} subcommands, and a /permissions slash command via internal/commands (registered through builtin/register.go). Followed F01's "extend existing" pattern. Engine proven via 3 integration tests + 3 Challenge scenarios; dispatcher wiring (ConfirmationCoordinator → permissions.Engine) deferred to Phase 3.
- 2026-05-05 — Feature 3 (Tool Result Persistence) closed. Eleven sub-commits. New thin sub-package internal/tools/persistence mirrors F02's pattern. Threshold check fires at the LLM provider boundary (tool_provider.go). T07 audit confirmed 0 of 16 LLM providers bypass tool_provider.go — single choke point. LLM reads back via the existing Read tool — no new tool added. Lazy 7-day CleanupOld at startup. Engine proven

via 3 integration tests + 3 Challenge scenarios (above/below threshold + hash idempotence).

- 2026-05-05 — Feature 4 (Git Worktree Agent Isolation) closed. Thirteen sub-commits (T02 needed a fix-up ccaaf33 for an anti-bluff smoke regression — the docstring contained “placeholders” which trips the coarse grep "placeholder" check). New thin sub-package `internal/tools/worktree` mirrors F02/F03’s pattern. Shells out to the git binary consistent with `internal/tools/git/`. Worktrees stored at `<repoRoot>/ .helix-worktrees/<name>/` (in-repo; .gitignore’d). Meta-only — no submodule auto-init; agents that need submodule code run `git submodule update --init --recursive` from inside the worktree. Full surface: 4 agent tools + 4 Cobra subcommands (enter/exit print help when called from CLI) + 1 /worktree slash command. Per-session state via single field on `session.Manager` rather than a parallel `worktree_state.go` file.
- 2026-05-05 — Feature 5 (Hook-Based Extensibility) closed. 14 sub-commits (12 feat + 2 fix-ups: T04’s cross-platform shell-runner split, T03’s yaml-loader priority default). Extended existing `internal/hooks` package with 6 new `HookType` constants + 3 new files (`yaml_loader`, `shell_runner`, `blockers`). Config-driven shell hooks via `~/.helixcode/hooks.yaml`. 5 wiring points: `tools/registry.Execute` (6 events), `llm/compression/AutoCompactor` (`OnCompaction`), agent (`OnError` + `RequestPlanApproval` stub for F08). Full surface: 5 Cobra subcommands + /hooks slash command (aliased /hk).
- 2026-05-05 — Feature 11 (Session Transcript Resume) closed. 9 task commits (T01 ddb45dc, T02 fa6bc5f, T03 466ab97, T04 d72e401, T05 08fa5c0, T06 607206a, T07 0fb036c, T08 submodule 1e79453 + meta f4d0ff2, T09 close-out) + 1 follow-up (f258cf7 preserves `ProjectPath/Name` across `SessionManager.Append`). New files in `internal/session/`: `identity.go` (Git-root-or-cwd), `transcript_store.go` (JSONL transcripts + metadata I/O), `resume.go` (`ResumeFinder` + `ResumeMode` + `FindResumeTarget`). Existing `session_manager.go` extended with `Append/Resume/CurrentID`. Surface: /sessions slash + `helixcode sessions {list,show,resume,delete} cobra --resume --continue` flags wired in `cmd/cli/main.go`. Challenge harness exercises real fork-exec process boundaries (write child PID ≠ read child PID, both ≠ orchestrator). All 4 remotes pushed non-force. F12 (Multi-Provider Backend) is the next candidate per the original 12-feature programme plan.
- 2026-05-05 — Feature 12 (Multi-Provider Backend) closed. 11 task commits (T01 bd5dc69, T02 06c9c34, T03 dde10dd, T04 d01026d, T05 67417ed, T06 880b4dc, T07 28b6fa1, T08 778040e, T09 ac55fca, T10 submodule 4e42fbc + meta-repo b937e17 + SHA backfill 1586624, T11 close-out). Anthropic / Bedrock / Vertex / Azure unified behind a single `Selector` (flag > env > config > wizard precedence) with verifier-backed `GetModels` on all four cloud providers (CONST-036/037 satisfied). New `internal/llm/{selector.go, wizard.go, wizard_writer.go, provider_factory.go::NewCloudProvider}` plus four per-provider audit tests (`{anthropic,bedrock,vertexai,azure}_provider_audit_test.go`). Surface: --provider flag + `HELIX_LLM_PROVIDER` env + `helixcode wizard cobra + tview` TUI wizard with `tcell.SimulationScreen` headless tests (mode 0600 + `O_EXCL` writer). Challenge harness emits LOCAL-always-runs (11/11 PASS) + CLOUD credential-gated sections with explicit SKIP-OK: P1-F12 cloud creds not provided markers. All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F13 (LSP Integration) is the next candidate.
- 2026-05-05 — Feature 13 (LSP Integration) closed. 12 task commits (T01 df98b6d, T02 b9a30e4, T03 3c5d894, T04 2fdb648, T05 beef346, T06 33387a3, T07 9bb3118, T08 a1aa7e6, T09 1b7812f, T10 080b79b, T11 submodule f00bf19 + meta-repo 9ea2cdf, T12 close-out). 5 curated LSP servers (`gopls` / `rust-analyzer` / `pyright` / `typescript-language-server` / `clangd`) brought online with lazy-spawn + 5-minute

idle timeout per server, file-extension router, crash recovery, post-Execute auto-trigger after `fs_edit/fs_write/multi_edit_commit` (attaches `lsp_diagnostics` to the tool result map). New files in `internal/tools/`: `lsp_types.go`, `lsp_client.go` (jsonrpc2 wrapper around `go.lsp.dev/jsonrpc2 v0.10.0` + `go.lsp.dev/protocol v0.12.0`), `lsp_manager.go`, `lsp_servers.go`, `lsp.go` (LSPGetDiagnostics + LSPAnalyzeDiagnostic agent tools), `lsp_autotrigger.go`, `lsp_fakeserver/main.go` (in-tree real-subprocess fake LSP for deterministic tests). Surface: `/lsp slash + helixcode lsp {status, restart, list-servers, stop}` cobra. Challenge harness has TWO sections — MANAGER PIPELINE (always runs, real OS subprocess speaking real LSP-framed JSON-RPC over stdio, NOT an in-process stub) + REAL LANGUAGE SERVERS (gated, SKIP-OK with install hints). All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F14 (Sandboxed Shell Execution) is the next candidate.

- 2026-05-06 — Feature 15 (Subagent Team) closed. 12 task commits (T01 b970aa5, T02 adc273d, T03 ceeb670, T04 ec21b17, T05 8e2f9e8, T06 9311692, T07 1f9d0f3, T08 07863d2, T09 87b6eac, T10 af0aa29, T11 submodule 163965e + meta-repo 16708a7, T12 close-out). Hybrid in-process + subprocess subagent dispatch with optional F04 worktree isolation: `internal/agent/subagent/` package (types + FakeLLMProvider TEST PROVIDER + InProcessSpawner + SubprocessSpawner + SubagentManager with streaming aggregation + max-concurrency + kill-by-id + helper_mode IsSubagentInvocation/RunAsSubagent + worktree_integration with CreateWorktreeForSubagent non-mutating helper on F04 manager). New task tool (claude-code-compatible name) at `internal/tools/task_tool.go`; new `/subagents slash` command (list/status/kill) at `internal/commands/subagents_command.go` with CONST-042 anti-leak (status shows description, never prompt body). Subagent recursion guard: subprocess child's RunAsSubagent does NOT register the task tool, capping subagent depth at 1 in v1. Three-line main.go integration: `subagent.IsSubagentInvocation()` early-main check (FIRST statement, before `sandbox.IsHelperInvocation()`), manager construction, slash registration. Challenge harness emits SIX phases — IN-PROCESS (always, asserts `GenerateCallCount==1`), SUBPROCESS (always, real fork-exec of harness binary, parent provider call count == 0), WORKTREE (gated on git, real `git init` + F04 worktree + staged diff), REAL-LLM (gated on ANTHROPIC_API_KEY), CONCURRENCY-CAP (always, 3rd Dispatch returns `ErrMaxConcurrency`), KILL-CANCEL (always, Kill propagates → `State=StateCanceled`). On this host phases A/B/C/E/F all RAN and PASS; phase D SKIPPED (no API key). All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F16 (OpenTelemetry Integration) is the next candidate.
- 2026-05-06 — Feature 16 (OpenTelemetry Integration) closed. 12 task commits (T01 5fc7dc1, T02 f2e7260, T03 de941b4, T04 3c8593c, T05 a8e13e3, T06 6fcbff6, T07 d80c278, T08 7c06806, T09 7701c33, T10 a5eb1c9, T11 submodule af34a2c + meta-repo c4972dc, T12 close-out). OTel v1.30.0 tracing + metrics with three exporters (OTLP/gRPC, OTLP/HTTP, stdout) + no-op fast path; env-var configured (`OTEL_`); *TracedLLMProvider decorator (Go struct embedding) wraps `llm.Provider Generate/GenerateStream`; in-place 5-line wraps in `internal/tools/registry.go::Execute` + `internal/agent/base_agent.go::executeTaskWithLLM`; pre-built instruments (`helixcode_llm_tokens_total` + `llm_latency_seconds` + `tool_calls_total` + `tool_latency_seconds` + `agent_iterations_total` + `agent_iteration_duration_seconds`); `DefaultBlockedAttributeKeys` covers credential keys (`api_key/token/bearer/password/secret/authorization/{anthropic,openai}api_key/aws`) + prompt-body keys (`prompt/prompt_body/request_body/response_body`) — case-insensitive `FilterAttributes` is default-deny; `TracedLLMProvider` NEVER emits prompt*

body as span attribute (CONST-042). Surface: `/telemetry {status, show, flush}` slash command (status renders KIND/ENDPOINT/SERVICE/SPANS/METRICS table; show prints stdout ring buffer; flush calls ForceFlush). Challenge harness emits FIVE phases — STDOUT (always, captures span+metric writers), FAKE-OTLP-HTTP (always, in-tree net/http. Server decoding real OTLP protobuf bodies), FILTER (always, deliberately injects `api_key=sk-CHALLENGE-12345` and asserts marker absent in export), NOOP (always, 100 calls with disabled telemetry), REAL-COLLECTOR (gated on `OTEL_EXPORTER_OTLP_ENDPOINT`). On this host phases A/B/C/D RAN and PASS; phase E SKIPPED (no collector). All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F17 (Smart File Editing) is the next candidate.

- 2026-05-06 — Feature 14 (Sandboxed Shell Execution) closed. 12 task commits (T01 0ef5811, T02 abdbdab, T03 4f7141f, T04 ec4cb9b, T05 5d05b3d, T06 a642101, T07 ba54c0c, T08 9aadc02, T09 93dc377, T10 fdb5ddc, T11 submodule 7d336ad + meta-repo 998896c, T12 close-out). Linux-first sandboxed shell execution: hybrid bubblewrap (preferred when on PATH) + native Go Cloneflags users fallback; default-DENY network with per-call opt-in; CONST-033 power-management deny-list rejects matching commands BEFORE any subprocess spawns; fail-closed when neither bwrap nor unprivileged users are available — never a silent unsandboxed run. New `internal/tools/sandbox/` package (types + detector + bubblewrap_backend + native_backend + native_helper re-exec + manager + sandboxed_shell_tool + config_loader). Tool: `shell_sandboxed`. Surface: `/sandbox {status, test, policy}` slash + secret-safe YAML at `~/ .config/helixcode/sandbox.yaml` (mode 0600, parent 0700 — mirrors F12 wizard_writer; CONST-042 satisfied). Challenge harness emits THREE sections — DETECTOR + FAIL-CLOSED (always runs, asserts CONST-033 spawn-counter rejection AND verbatim fail-closed message), BUBBLEWRAP (gated, real curl-inside-sandbox network probe), NATIVE (gated, force-constructed NativeBackend with helper-mode dispatch). On this host both gated phases RAN end-to-end. All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F15 (Subagent Team) is the next candidate.
- 2026-05-06 — Feature 17 (Smart File Editing) closed. 10 task commits (T01 37b1471, T02 adcd9f0, T03 91d7550, T04 37beb27, T05 6c00471, T06 721fed9, T07 a2dd7eb, T08 5bf4c92, T09 submodule e2e9e94 + meta-repo daa1279, T10 close-out). Aider-style SEARCH/REPLACE block edits with multiedit-transactional atomicity. New `internal/tools/smartedit/` package (types + parser + applier + binary_detect + diff + smart_edit_tool) wrapping F08's `multiedit.MultiFileEditor` for atomic multi-file commits with post-write re-read verification. Lenient re-search across consecutive blocks; ambiguity rejection (multiple matches → fail-closed); binary-file refusal (NUL byte detection in first 8KB); pure-Go unified-diff via `multiedit.DiffManager.GeneratedDiff` byte-exact-matching system `diff -u`. Surface: `/edit {status, diff, dry-run, commit}` slash command + tool registry registration as `smart_edit`. Challenge harness emits SEVEN phases — SINGLE-FILE-SUCCESS, SEARCH-NOT-FOUND-REJECTED, MULTI-FILE-ATOMIC, PARTIAL-FAILURE-ROLLBACK (chmod 0500 on file 3's parent dir; sha-256 byte-identity assertion across all 5 files), DIFF-EXACTNESS (byte-exact match vs system `diff -u`), AMBIGUOUS-REJECTED, BINARY-REFUSED — each with sha-256 positive evidence, never absence-of-error. All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F18 (No-Flicker Rendering) is the next candidate.
- 2026-05-06 — Feature 19 (AskUserQuestion with Previews) closed. 7 task commits (T01 b7fc6bb, T02 3be8ca7, T03 2ded9d2, T04 bc84789, T05 98bd424, T06 submodule e3454a4 + meta-repo ecb3e26, T07 close-out). Real, end-to-end `ask_user` tool for the HelixCode CLI agent: structured multiple-choice prompt with

optional inline previews. New `internal/tools/askuser/` package: `types.go` (Choice + Question + Result + Prompter interface + sentinels `ErrInvalidQuestion/ErrNoTTYNoDefault/ErrUserCancelled/ErrPromptTimeout/ErrTooManyInvalidAttempts` + constants `DefaultTimeout=5min/DefaultMaxRetries=3/SourceStdin/SourceDefault`), `stdin_prompter.go` (production `stdinPrompter` with `Reader/Writer/IsTTY/Renderer/Timeout/MaxRetries` constructor seams; non-TTY short-circuit BEFORE any read; TTY branch via `bufio.ReadString('\n')` in goroutine + `select-on-ctx.Done`; retry loop on empty/non-numeric/out-of-range; EOF → `ErrUserCancelled`; `ctx-cancel/timeout` → `ErrPromptTimeout`; F18-renderer reused for prompt menu + per-choice inline preview), `ask_user_tool.go` (`AskUserTool` implements `tools.Tool`; `Name()=="ask_user"`; `Category()==tools.CategoryAskUser`; returns `map[string]interface{}{"value","label","index","source"}`). Two existing files extended: `internal/tools/registry.go` (new `CategoryAskUser` const + `AskUserPrompter` field on `RegistryConfig` + registration in `buildToolList`); `cmd/cli/main.go` (no changes required for correctness). Zero new external deps — pure `stdlib` + `golang.org/x/term` (already direct dep after F18) + `dev.helix.code/internal/render`. Integration tests (`tests/integration/askuser_test.go`, build-tag integration) exercise the production `stdinPrompter` end-to-end through `AskUserTool` with real bytes. Buffer reader/writer. Challenge harness emits FIVE always-run phases — PHASE-A (TTY-WITH-INPUT-RETURNS-CHOICE), PHASE-B (NON-TTY-WITH-DEFAULT, asserts `reader-remaining=2` untouched=`true` + `writer-bytes=0`, proving short-circuit before read), PHASE-C (NON-TTY-NO-DEFAULT-ERRORS, asserts `errors.Is(ErrInteractiveTerminalRequired)=true` + `writer` untouched), PHASE-D (PREVIEW-VISIBLE-IN-OUTPUT, asserts byte offsets 37 + 87 of preview text inside writer — preview BEFORE label by byte order), PHASE-E (INVALID-INPUT-RETRY, retry loop reads new bytes, question rendered 2× with “1-3” hint). Cross-compile `linux/amd64` produces a 73 MB binary. Anti-bluff smoke clean across `internal/tools/askuser/`, `cmd/cli/main.go`, `internal/tools/registry.go`, `tests/integration/askuser_test.go`, `tests/integration/cmd/p1f19_challenge/`, and `challenges/p1-f19-ask-user-question/`. All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). F20 (Theme System) is the next candidate per the porting doc.

- 2026-05-06 — Feature 18 (No-Flicker Rendering) closed. 10 task commits (T01 a8aa8f3, T02 8d6ec3b, T03 a3cd0e1, T04 487d72e, T05 8c90e7c, T06 288e6cd, T07 4ece7e8, T08 05434c4, T09 submodule c409ed3 + meta-repo c44b049, T10 close-out). Custom ANSI/CR renderer for the HelixCode CLI streaming hot path with non-TTY plain fallback; zero new external dependencies (pure `stdlib` + `golang.org/x/term` promoted indirect→direct). New `internal/render/` package: `types.go` (`Renderer` interface + `RenderMode` + `Frame` + sentinels + env-var consts), `ansi_renderer.go` (in-place line update via `\r\x1b[K + \x1b[?25l/h` cursor hide-show + dirty-region multi-line frame diff via cursor-up/down), `plain_renderer.go` (line-by-line `fmt.Fprintln + \r-strip` + zero-ANSI/zero-CR invariants + ANSI pass-through per §5.4), `viewport.go` (pure-Go dirty-line Diff + Apply), `factory.go` (`HELIXCODE_RENDER=plain|fancy|auto` env + `term.IsTerminal` TTY detection + constructor-injection seams), `tool_helpers.go` (`RenderTextBlock/RenderLines/RenderToolResult` glue with type-switch over `Frame / []string / string / Stringer / fallback`). Surface: env-var only (no slash, no cobra subcommand — Q5=B). Wired into `cmd/cli/main.go`: renderer constructed at startup, `defer renderer.Close()`, streaming-print loop in `handleGenerate` replaced with `Begin/WriteToken/Commit`, non-streaming

Generate response printed via `RenderTextBlock`. Challenge harness emits FIVE phases — STREAMING-FANCY (10 `\r\x1b[K` sequences + `\x1b[?25l/h` pair, 371 bytes), STREAMING-PLAIN (zero `\x1b` and zero `\x0d`, 58 bytes), DIRTY-REGION-DIFF (one-line-changed `delta=34 < firstLen=80 + exactly 1 cursor-up`), TTY-FALLBACK (factory ladder picks plain on `bytes.Buffer` non-TTY), REAL-TTY (gated; SKIP-OK on non-TTY) — each with positive byte evidence, never absence-of-error. All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its `origin` (mirror gap noted, deferred infra). F19 (`AskUserQuestion` with Previews) is the next candidate.

- 2026-05-06 — Feature 20 (Theme System) closed. 9 task commits (T01 60777fd, T02 4697565, T03 a66737d, T04 7f97f57, T05 1fd42d9, T06 1066798, T07 348630c, T08 submodule 4bf04bb + meta-repo 300f973, T09 close-out). Real, end-to-end 5-role semantic theme system for the HelixCode CLI agent: built-in dark / light / none themes with byte-pinned palettes per spec §3.4 across three depth tiers (Truecolor / ANSI256 / ANSI16); optional YAML override at `$XDG_CONFIG_HOME/helixcode/theme.yaml` slot-merged over the dark baseline; theme-name resolution ladder (`HELIXCODE_THEME` env > `$COLORFGBG` heuristic > dark default); color-depth auto-detect from `$COLORTERM` (truecolor/24bit) and `$TERM` (*-256color → ANSI256, dumb/empty → Off, otherwise ANSI16); Styler decorator wraps text with role-coded SGR sequences (decorator pattern over F18's `Renderer` — F18 interface unchanged); plain-mode forced to `DepthOff` via `WithDepth` at startup as belt-and-suspenders against renderer regressions; `/theme {status,list,show <name>}` slash command (read-only). Zero new external deps — `gopkg.in/yaml.v3` was already present in `go.sum` via viper/cobra. Five always-run Challenge phases — PHASE-A (BUILT-IN-DARK, 5/5 dark roles with pinned `\x1b[38;2;...m` truecolor opens + `\x1b[0m` Reset), PHASE-B (BUILT-IN-LIGHT, light error `\x1b[38;2;175;0;0m` ≠ dark error `\x1b[38;2;255;64;64m` cross-theme distinguisher), PHASE-C (PLAIN-ZERO-COLOR, `bytes.IndexByte == -1` invariant under `DepthOff`), PHASE-D (DEPTH-DETECT, 6 env-driven branches), PHASE-E (YAML-MERGE, real `theme.yaml` on `tmpdir` parsed by `yaml.v3`, custom error slot overlays dark baseline while un-mentioned `info/warn/highlight/dim` inherit). All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its `origin`. **PHASE 1 OF CLI-AGENT FUSION PROGRAMME COMPLETE — 20 features (F01..F20) shipped.**
- 2026-05-06 — **PHASE 1 OF CLI-AGENT FUSION PROGRAMME COMPLETE.** All 20 features shipped: F01 Auto-Compaction (4734f35+9284392 evidence; close-out f0b9b15.); F02 Permission Rules (d56905d..close-out); F03 Tool Result Persistence (ee35017..close-out); F04 Git Worktree Agent Isolation (d5ae14a..close-out); F05 Hook-Based Extensibility (b7e7185..close-out); F06 MCP Full Lifecycle (T14 close-out); F07 Background Tasks (T11 close-out); F08 Plan Mode (T09 close-out); F09 Slash Commands (T08 close-out); F10 Skills (T09 close-out); F11 Session Resume (ddb45dc..T09 close-out); F12 Multi-Provider Backend (bd5dc69..T11 close-out); F13 LSP Integration (df98b6d..T12 close-out); F14 Sandboxed Shell Execution (0ef5811..T12 close-out); F15 Subagent Team (b970aa5..T12 close-out); F16 OpenTelemetry (5fc7dc1..T12 close-out); F17 Smart File Editing (37b1471..T10 close-out); F18 No-Flicker Rendering (a8aa8f3..92db29e close-out); F19 `AskUserQuestion` (b7fc6bb..f584c67 close-out); F20 Theme System (60777fd..T09 close-out — this commit). Phase 1 entry condition met: every Phase 1 feature ships with positive runtime evidence per Article XI §11.9 (no absence-of-error PASS), all anti-bluff smoke clean, all four meta-repo remotes (`origin + github + gitlab + upstream`) at parity, all Challenges submodule changes mirrored to its `origin`. Next phase: Phase 2 (other CLI agents) once user authorises.
- 2026-05-06 — **PHASE 1.5 (FOUNDATION CLEANUP) COMPLETE.** All 12 work packages shipped: WP1 Inventory + foundation safety (421495a); WP2 Submodule restructuring ~67 mechanical moves (90dec95); WP3 Submodule deduplication 5 sets

(154c06c); WP4 API key loader bash + Go (e57894e); WP5 .env API key dedup with USER GATE (92d5463); WP6 Docs consolidation 3 dirs (f09f57d); WP7 Snake_case directory normalization (3c3cd8d); WP8 Anti-bluff Constitution propagation (0eead08); WP9 Reference updates comprehensive grep sweep (42166fd); WP10 Rebuild + validation + fix internal/tools/git MockLLMProvider drift (0a77c93 + fix 45be827); WP11 Phase 1.5 Challenge harness 5 phases (meta 306d3d9 + Challenges submodule 7e94f28); WP12 close-out + push deepest-first (this commit). Acceptance criteria: Phase 1.5 Challenge harness EXIT=0 with all 5 phases (NO-DUPLICATE-SUBMODULES + API-KEYS-LOADER + DOCS-UNDER-DOCS-DIR + SNAKE_CASE + ANTI-BLUFF-ANCHOR) printing positive runtime evidence per Article XI §11.9; scripts/verify_anti_bluff_cascade.sh exit 0 (“OK: anti-bluff anchor present in all 39 files across 13 repos”); inner-module go test -count=1 -short ./internal/... ./cmd/... all green; anti-bluff smoke clean across all P1.5-touched files. All four meta-repo remotes (origin + github + gitlab + upstream) at parity; HelixAgent submodule pushed to its origin; Challenges submodule pushed to its origin. **Next phase: Phase 2 (other CLI agents) ready to start once user authorises.**

- 2026-05-06 — Phase 2 started; F21 (Codex Approval Modes) is first port; layers on F02 + F14; uses F19 prompter; CLI/env/config selector mirrors F12 pattern.
- 2026-05-06 — Feature 21 (Codex Approval Modes) closed. 9 task commits (T01 a7a349f, T02 9c2664d, T03 0d655d8, T04 5ef13b8, T05 19bfffce + CONTINUATION 1195ef9, T06 ad8843b + CONTINUATION 9b72c26, T07 c022968 + CONTINUATION bd67324, T08 Challenges submodule f2ea964 + meta-repo 2781c1a + CONTINUATION ee413c3, T09 close-out — this commit). Real, end-to-end 4-mode codex-compatible approval system: suggest (read-only) / auto-edit (edit OK, run prompts) / full-auto (edit + run with sandbox forced + network DENY) / dangerously-bypass (no checks; 2-second pause + warning). Selector: -approval flag > HELIXCODE_APPROVAL env > ~/.config/helixcode/approval.yaml > default suggest. Per-tool Tool.RequiresApproval() ApprovalLevel (read-only/edit/run/all); ~38 existing tools migrated with explicit levels; safe-default LevelEdit for forgotten overrides. /approval slash (status/set/show); runtime mode swap via atomic.Pointer takes effect on next CheckApproval. F02 retains final-deny authority (composition truth table pinned via instub deny-rule in PHASE-E pending registry-level seam). Zero new external deps. Five-phase Challenge harness PASS (PHASE-A SUGGEST-DENY + PHASE-B AUTO-EDIT-PROMPT + PHASE-C FULL-AUTO-SANDBOX + PHASE-D RUNTIME-CHANGE + PHASE-E F02-FINAL-DENY) with positive runtime evidence per Article XI §11.9; cross-compile linux/amd64 PASS (94 MB binary); anti-bluff smoke clean across all F21 surface (HelixCode + Challenges). All 4 meta-repo remotes pushed non-force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). **First Phase 2 feature shipped.** F22 next candidate per synthesis design §4.2 (Codex follow-on / aider / cline / plandex — choice via brainstorming).
- 2026-05-07 — Feature 24 (Codex Project Memory) closed. 8 task commits (T01 f55b3e3, T02 fd90eed, T03 99d3971, T04 0760562, T05 d740964, T06 7af2859, T07 40927fc, T08 close-out — this commit). Real, end-to-end project memory subsystem modelled on codex’s AGENTS.md pattern. NEW internal/projectmemory/ package: Memory (immutable value with Render() = Project + delimiter + User) + MemoryLoader (parent-walk discovery for helixcode.md / codex.md / AGENTS.md, case-insensitive two-pass scan, stops at git-root marker or filesystem root, real tempdir-tested) + MemoryRegistry (atomic.Pointer[Memory] Snapshot lock-free / Set / Reload mu-serialised + MemorySnapshotter interface) + MemoryWatcher (real fsnotify v1.9.0 + 200 ms debounce + parent-dir watch survives atomic-write rename + graceful degrade on watch failure). NEW /memory slash (status/show/edit/reload) with \$EDITOR fallback to vi. BaseAgent.SetMemoryRegistry + getSystemPrompt prepends Memory.Render() per-call (lock-free Snapshot —

NOT cached at construct time; spec §5.2 Bluff #3). User overlay at `$XDG_CONFIG_HOME/helixcode/memory.md` loaded AFTER project memory (security invariant). 64 KB cap with positive `TruncatedProject / TruncatedUser` flags. Composes with F15 subagents (shared registry per CLI-instance), F04 worktrees (per-cwd discovery), F21 approval / F22 autocommit / F23 browser (orthogonal — registry is pre-existing main.go state). Zero new external deps (`fsnotify v1.9.0` already direct in `helixcode/go.mod`; `go.mod + go.sum` diff-clean post `go mod tidy`). Five-phase Challenge harness PASS (PHASE-A PROJECT-ONLY 4/4 + PHASE-B MISSING-FILE-GRACEFUL 4/4 + PHASE-C HOT-RELOAD 3/3 + PHASE-D PROJECT-PLUS-USER 3/3 + PHASE-E TRUNCATION 3/3 = 17/17 checks) with positive runtime evidence per Article XI §11.9 against real tempdirs + real `fsnotify`; cross-compile linux/amd64 PASS (45 MB binary); anti-bluff smoke clean across all F24 surface (HelixCode + Challenges); unit/integration tests all green under `-race`. **Fourth Phase 2 feature shipped.** F25 next candidate per synthesis design §4.2.

- 2026-05-07 — Feature 23 (Cline Browser Tool) closed. 10 task commits (T01 64e499b, T02 cdb323e, T03 e0ff1bf, T04 2bcb281, T05 ec0b3cc, T06 659307d, T07 3ca50a3, T08 ad0c0df, T09 f39f686, T10 close-out — this commit). Real, end-to-end 6-tool browser-automation suite modelled on cline: `browser_navigate / browser_snapshot / browser_click / browser_type / browser_screenshot / browser_close`. Atomic-pointer `BrowserManager`; lazy-create on first navigate; `sync.Once` close. Headless default; `HELIXCODE_BROWSER_HEADED=true` opt-in. Screenshots written to `$XDG_DATA_HOME/helixcode/browser/screenshots//.png` with PNG-magic + `DecodeConfig` + `size>1024` verification. F21 `RequiresApproval` per-tool: `navigate/click/type/close` → `LevelEdit`; `snapshot/screenshot` → `LevelReadOnly`. `/browser slash (status/navigate/close)`. NO cobra. Coexists with renamed legacy `browser_legacy_*` multi-browser tools. Composes with F21 approval (gates `LevelEdit` tools), F22 autocommit (no spurious commits — browser tools have no path param), F13 LSP trigger (no-op for browser tools), F04 worktree (browser per CLI-instance). Zero new external deps (`chromedp v0.15.1` + `cdproto` already direct). Seven-phase Challenge harness PASS (PHASE-A NAVIGATE-AND-SNAPSHOT 4/4 + PHASE-B SNAPSHOT-MODE-TEXT 2/2 + PHASE-C CLICK-MUTATES-DOM 3/3 + PHASE-D TYPE-INTO-INPUT 2/2 + PHASE-E SCREENSHOT-PNG-MAGIC 4/4 + PHASE-F CLOSE-TEARS-DOWN 2/2 + PHASE-G CONCURRENT-SESSION-SHARING 2/2) with positive runtime evidence per Article XI §11.9 against real chromium subprocess; cross-compile linux/amd64 PASS (45 MB binary); anti-bluff smoke clean across all F23 surface (HelixCode + Challenges); 7/7 integration tests PASS (3.737s) against real chromium. **Third Phase 2 feature shipped.** F24 next candidate per synthesis design §4.2.
- 2026-05-07 — Feature 22 (Aider Git Auto-Commit Per Change) closed. 9 task commits (T01 550be34, T02 0468beb, T03 cb4fc30, T04 4b2ab67, T05 3a28ca6, T06 db55e72, T07 a999b3a, T08 bab7ebc, T09 close-out — this commit). Real, end-to-end per-edit git auto-commit modelled on aider: one commit per accepted edit (`LevelEdit/LevelAll` tools); LLM-summarised subject (50-72 chars, imperative voice) with deterministic fallback on LLM unavailability; `Co-Authored-By: HelixCode <noreply@helixcode.dev>` trailer appended to EVERY auto-commit. Default-on; opt-out via `HELIXCODE_GIT_AUTO_COMMIT=off` env, runtime / `git_auto_commit off slash`, or per-edit `_helix_skip_git_commit:true` param. Composes with F21 approval (only commits AFTER approval granted) and F04 worktree (auto-commits work in subagent worktrees). Best-effort secret filter (CONST-042). Zero new external deps. Seven-phase Challenge harness PASS (PHASE-A DEFAULT-ON 7/7 + PHASE-B LLM-SENTINEL 3/3 + PHASE-C NON-EDIT-NO-OP 2/2 + PHASE-D ENV-OFF 3/3 + PHASE-E RUNTIME-TOGGLE 4/4 + PHASE-F PER-EDIT-SKIP 3/3 + PHASE-G SECRET-FILTER 3/3) with positive runtime evidence per Article XI §11.9; cross-compile linux/amd64 PASS (44 MB binary); anti-bluff smoke clean across all F22 surface (HelixCode + Challenges). All 4 meta-repo remotes pushed non-

force; Challenges submodule pushed to its single origin (mirror gap noted, deferred infra). **Second Phase 2 feature shipped.** F23 next candidate per synthesis design §4.2 (cline / plandex / openhands — choice via brainstorming).

Open risks / parking lot

- **Historical SSH key leak (remediated in P0-T08.5):** `id_rsa + id_rsa.pub` at `helix_code/test/workers/ssh_keys/` were committed as test fixtures before this programme. Their material lives in git history forever and is considered compromised. Mitigation: keys were ephemerally test-only (no production trust), replaced with auto-generated ed25519 ephemeral keys via `helix_code/test/workers/ssh_keys/generate-test-keys.sh`, removed from the index via `git rm --cached`. Any future production system that erroneously trusts the leaked public key must reject it.
- **Historical helix.security.json credential leak (remediated in P0-T08.5):** `helix.security.json` at repo root was committed with real SonarQube and Snyk credentials (token, project_key, organization, url). Material lives in git history and is considered compromised. Mitigation: removed from index via `git rm --cached`; replaced with `helix.security.json.example` containing `<REDACTED>` placeholders. Rotate: SonarQube token, Snyk token, organization, and project_key immediately.
- HelixAgent submodule clone size — may need `--depth=1` shallow if >500 MB; measured at P0-03
- Codex agent disambiguation (closed vs. open variant) — deferred to Phase 2 sub-spec
- Example_Projects/ deletion — deferred to post-Phase-4 decision
- **Submodule recursion cosmetic error (deferred from P0-02):** `git submodule foreach --recursive` errors out on `Example_Projects/{Agent-Deck,Bridle,Claude-Code-Plugins-And-Skills}` because each of those third-party repos has registered nested gitlinks (mode 160000) without corresponding `.gitmodules` entries. The original Task 2 plan proposed `.git/info/exclude` — that does NOT fix recursion (which walks the index, not the working tree). The only safe in-scope fix is to wrap script calls with `|| true` and tolerate the error. Modifying the affected third-party submodules' contents is forbidden by spec §2.1 (third-party not modified). Decision: scripts that use `git submodule foreach --recursive` (none yet in our codebase) must wrap with `|| true`; documentation updates that erroneously claimed Task 2 would resolve this are corrected.
- **HelixAgent stale cli_agents pins (discovered during P0-03):** 13 of 60 `cli_agents` under `helix_agent/cli_agents/` cannot be initialized because HelixAgent's recorded submodule SHAs no longer exist on the corresponding upstream remotes. Affected: `aider`, `conduit`, `continue`, `HelixCode`, `kilo-code`, `kiro-cli`, `mobile-agent`, `ollama-code`, `opencode-cli`, `openhands`, `plandex`, `roo-code`, `superset`. Each Phase 2 sub-spec for the affected agent must first bump HelixAgent's pointer (commit IN HelixAgent itself, then bump HelixAgent's pointer in this meta-repo) to a SHA that exists upstream. Phase 1 priority `claude-code` is NOT affected — fully populated. Per spec §1.3 N2, HelixAgent rewrite is out of scope for this programme; the per-agent pin bumps go through HelixAgent's own governance.
- **SonarQube + Snyk live-scan deferral (P0-T08.7):** The scan infrastructure (compose files, scripts, BootManager binary, Challenges) is fully wired and configuration-validated. Live scans CANNOT run until the user rotates the leaked credentials from `helix.security.json` (remediated in P0-T08.5 but historical values are compromised). Action required: (1) generate new SonarQube API token, (2) set `SONAR_TOKEN + SONARQUBE_PROJECT_KEY + SONARQUBE_PROJECT_NAME` in `helix_code/.env`, (3) generate new Snyk token, (4) set `SNYK_TOKEN` in

helix_code/.env, (5) run make scan-sonarqube / make scan-snyk. This is NOT a code defect — it is a security-rotation dependency on the operator.

- **LLMsVerifier dual-pin divergence (discovered during P0-04):** RESOLVED in P1.5-WP2; doc previously stale. The transitive duplicate helix_agent/LLMsVerifier was eliminated in WP2 (only a backup remains at helix_agent/.container-caps-backup-20260425-151947/LLMsVerifier). helix_agent/go.mod:232 now replaces digital.vasic.llmsverifier => ../submodules/llms_verifier/llm-verifier, so HelixAgent transitively consumes the canonical pin. The pin-parity gate scripts/verify-llmsverifier-pin-parity.sh is degenerate (regression-tripwire only; header confirms WP2 elimination). make verify-foundation evidence (close-out⁴⁵, 2026-05-15): exits 0, “0 failures, all gates passed” — no VERIFY_FOUNDATION_WARN_ONLY waiver required. Original divergence text preserved verbatim above this annotation in the historical evidence at 05_phase_0_evidence.md §P0-04.
- **Permissions engine not yet threaded into tool dispatch (deferred from P1-F02-T09):** The --permission-mode flag parses and permissions.Engine constructs at startup, but the resulting Engine’s *confirmation.PolicyEngine is currently local to (*CLI).initPermissions and is not consulted by the production tool-execution path. The engine itself is proven correct (3 integration tests + 3 Challenge scenarios); the wiring gap means a deny rule would not actually block a tool call in a live session. Action: Phase 3 (test infra) sub-spec must wire internal/tools/registry.go’s ConfirmationCoordinator to use the loaded engine. Current behavior: rule files are validated and the CLI flag is honored at the helixcode permissions check dry-run; live tool dispatch falls through to the default confirmation.PolicyEngine (which has no rules). NOT a security regression — falls back to ask-by-default.

P2-F28 task list (Kilo-code Refactoring) — ALL CLOSED



P2-F28-T01 — bootstrap F28 evidence + advance PROGRESS



P2-F28-T02 — types.go + callgraph.go: CallGraph + BuildCallGraph (TDD)



P2-F28-T03 — rename.go: tree-sitter query + atomic F17 edits (TDD)



P2-F28-T04 — impact.go: ImpactAnalyzer + blast radius (TDD)



P2-F28-T05 — refactor.go: extract method, move, inline (TDD)



P2-F28-T06 — kilocode_tools.go: 3 tools + /kilocode slash (TDD)



P2-F28-T07 — main.go wiring



P2-F28-T08 — Challenge harness 5 phases + close-out + push 4 remotes

Phase 3 — Issue remediation (2026-05-07)

RESOLVED:

- **F23 browser_legacy_* approval test** — `TestAllRegisteredTools_DefaultIsLevelEdit_ForUnclassifiedTools` now PASSES.
- **F02 permissions engine wired** — `ConfirmationCoordinator.SetPolicyEngine()` method added. `main.go` now injects permissions-loaded `PolicyEngine` into the confirmation pipeline. Deny rules from `.yaml` files now actually block tool execution.
- **Challenges submodule mirrors** — `gitlab` and `upstream` remotes added to Challenges submodule.
- **Governance cascade** — `CONST-033/035/042/043/045` cascaded to Security and containers submodules.
- **LLMsVerifier build** — Fixed `go.mod` replace path `../../Challenges` → `../../../../Challenges` (`LLMsVerifier` at `submodules/llms_verifier/`, Challenges at meta-repo root `challenges/`). `go build ./...` and `go mod tidy` both pass clean.
- **HelixQA build** — Fixed 4 replace directives that pointed to wrong paths: `../LLMsVerifier/llm-verifier` → `../submodules/llms_verifier/llm-verifier`, `../DocProcessor` → `../submodules/doc_processor`, `../LLMOrchestrator` → `../submodules/llm_orchestrator`, `../VisionEngine` → `../submodules/vision_engine`. `go build ./...` and `go mod tidy` both pass clean.
- **6 internal/server tests** — Now 0 FAIL. All internal/server tests PASS (fresh `-count=1` run). Previously listed as “pre-existing, environment-dependent” but now passing in current environment.
- **HelixAgent submodules** — Auth, Cache, Concurrency, Database and 100+ submodules now populated via `git submodule update --force --init --recursive`. `go build ./cmd/helixagent/...` PASS. `go test ./cmd/helixagent/...` and `go test ./internal/...` both PASS. Only `DebateOrchestrator` (repo not on GitHub) blocks wildcard `./...`.
- **Containers build** — `go build ./...` exits 0. Build passes clean.

REMAINING (pre-existing, not Phase 2):

- **HelixAgent build** — IMPROVED: submodules now populated (Auth, Cache, Concurrency, Database all have `go.mod`). `go build ./cmd/helixagent/...` passes. `go test ./cmd/helixagent/...` and `go test ./internal/...` both PASS. Full `go test ./...` blocked by single stale replace directive: `digital.vasic.debate => ./DebateOrchestrator` (`DebateOrchestrator` repo doesn't exist on GitHub — pre-existing, HelixAgent-internal, out of scope per spec §1.3 N2).
- **Containers build** — RESOLVED: `go build ./...` exits 0. Build now passes clean.
- **Historical credential leaks** — operator rotation required
- **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream. Per spec §1.3 N2, HelixAgent rewrite is out of scope; per-agent pin bumps go through HelixAgent's own governance.
- **23 snake_case renames** — build-path-breaking, deferred
- **Codex Multimodal, Cline Computer Use** — not yet ported

Zero-Bluff Completion Programme — 5 Phases ALL CLOSED (2026-05-12)

Spec: docs/superpowers/specs/2026-05-08-helixcode-zero-bluff-completion-design.md

Phase	Title	Status	SHA
ZB-P1	Governance Propagation	DONE	5f9bb90
ZB-P2	Stub / Bluff Elimination (T01-T11)	DONE	b24ca8f
ZB-P3	Feature Gap Implementation (11 tasks)	DONE	1f1d8f4
ZB-P4	Test / Challenge Hardening (weak-assertion sweep)	DONE	21e6686
ZB-P5	Full Documentation Suite	DONE	(this commit)

ZB-P5 deliverables

Document	Path	Purpose
User Manual	docs/user_manual/ZERO_BLUFF_USER_MANUAL.md	End-user installation, configuration, F01-F30 usage
Developer Guide	docs/developer_guide/README.md	Architecture, Tool/Command/Skill interfaces, anti-bluff discipline
API Reference	docs/api_reference/README.md	20 key internal packages, content from go doc
Deployment Guide	docs/deployment_guide/README.md	containers submodule, observability, manual CI/CD, security

Anti-bluff smoke at close-out: clean (template placeholder matches in internal/config/config.go, internal/template/template.go, internal/workflow/*executor.go are legitimate domain usage referring to {{name}} template variable substitution — NOT TODO markers).

CLI examples validated against the codebase: - helixcode wizard — confirmed at helix_code/cmd/cli/wizard_cmd.go:83 - /approval, /memory, /lsp, /theme, /telemetry, /worktree — all present under helix_code/internal/commands/ - cli --list-models and interactive> models — confirmed at helix_code/cmd/cli/main.go:347, 1681 - cli sessions list — confirmed at

helix_code/cmd/cli/main.go:1822 - cli lsp list-servers — confirmed at
helix_code/cmd/cli/main.go:1850

Zero-Bluff Programme: COMPLETE

No further programme work pending. Backlog / parking-lot items: - cmd/infrastructure/
pre-existing compile errors against the containers submodule API — **RESOLVED 2026-05-12:**
go build ./cmd/infrastructure/... exits 0. Fixes applied: NewSlogAdapter() → NewSlogAdapter(nil); removed 16 WithDescription(...) calls (no longer on endpoint.Builder); migrated boot.Option → boot.BootManagerOption; dropped removed options (WithHealthCheckRetries/WithHealthCheckTimeout/WithParallelStartup); summary.Errors → walk summary.Results filtering by Status == "failed"; GetHealthStatus() → HealthCheckAll(ctx); removed unused time + compose imports; replaced ep.Description with ep.ServiceName in status output. - Fyne desktop build requires libxcursor-dev system package; documented as a prerequisite in the user manual.